

Raj Khatri

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PROFESSIONAL SUMMARY

Final-year Industrial & Production Engineering student working at the intersection of plant diagnostics and applied GenAI. At Tata Motors, diagnosed a live Body-in-White production defect by separating an upstream supplier-side cause from on-line equipment wear, using regression-based trend analysis on inspection data and confirming it by physical measurement on the shop floor. Built and deployed Source-Governed RAG, a retrieval system in which source approval status overrides semantic similarity — raising retrieval accuracy from 78% to 95% and forcing refusal rather than ungrounded generation when the corpus cannot support an answer. Reads plant process reasoning and writes Python.

EDUCATION

B.Tech, Industrial & Production Engineering
College of Technology, G.B. Pant University of Agriculture & Technology, Pantnagar

Uttarakhand, India
August 2023 – August 2027

TECHNICAL SKILLS

Languages:	Python, SQL (PostgreSQL / MySQL), JavaScript, Bash
GenAI & ML:	LLM application development, prompt engineering & constraint design, RAG pipelines, agentic tool routing, vector search (ChromaDB, pgvector), embeddings (MiniLM / ONNX), grounded generation & decline behaviour, evaluation-set design, Scikit-learn
Backend:	FastAPI, Pydantic data contracts, SQLAlchemy, PostgreSQL, pgvector, star-schema modelling, Flask, Next.js, Supabase, REST API design
Data & Tools:	Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Excel (Advanced / VBA), Power BI, Git/GitHub, Render, Vercel, Neon
Domain:	Root-cause analysis, upstream/downstream fault isolation, DFMEA, Six Sigma, statistical modeling & hypothesis testing, evaluation-set design

EXPERIENCE

Tata Motors <i>Industrial Trainee — Data Analytics</i> <i>Project: "Predictive Quality Intelligence — A Data-Driven Framework for BIW Root-Cause Diagnosis"</i> - Diagnosed a recurring Body-in-White quality failure as an upstream supplier-side dimensional defect, ruling out on-line equipment wear as the cause. - Applied linear regression with R² confidence scoring across ~300 inspection data points, then validated the result by physical measurement on the shop floor. - Designed a containment procedure deployable without production stoppage or vendor dependency. - Built an interactive diagnostic dashboard tracking defect trends across 10 monitored quality metrics for ongoing condition review. - Authored a 20+ section technical report benchmarking the findings against DFMEA, Six Sigma, and peer-reviewed manufacturing literature.	Pantnagar, Uttarakhand (On-site) 03 July 2026 – 03 August 2026
Voldebug Innovations Pvt. Ltd. <i>Backend Developer (Part-time)</i> - Migrating a live K–12 AI-literacy platform from browser-local persistence to a Supabase (PostgreSQL) backend — schema design, row-level security policies, and role-based authentication for student, teacher, and principal accounts. - Implementing server-side APIs in Next.js (App Router) for XP/badge progression, class-code enrollment, and teacher analytics, replacing client-only state with a shared data layer. - Establishing database migrations, staging/production environment separation, and a repeatable deployment workflow on Vercel.	Remote 22 August 2026 – Present
Intern - Built the platform's first working version — a gamified AI-literacy SPA spanning student, teacher, and principal roles, with a curriculum-linked prompt library and an XP/badge/streak progression system. - Authored the phased Supabase + Next.js migration plan now being implemented in the backend role above.	01 July 2026 – 31 July 2026

PROJECTS

Supplier Quality Risk Agent <i>Independent project · Python, FastAPI, Postgres + pgvector, Pydantic · Render / Vercel / Neon</i> - Built a dual-source AI agent that decides per query whether an answer requires structured supplier data, corrective-action documents, both, or neither — and declines when no source can support one — reaching 88–92% routing accuracy on a 24-question evaluation set. - Engineered a paired test case in which two suppliers show statistically identical defect-rate increases requiring opposite responses — one a genuine process degradation, one a measurement artefact — making single-source failure demonstrable rather than theoretical. - Replaced text-to-SQL with a fixed catalogue of eight parameterised queries, trading question coverage for reviewability in a system where a wrong number triggers a supplier corrective action. - Caught a regression where a prompt improvement made the router fabricate evidence for a non-existent supplier; constrained the prompt with the real entity list, raising decline accuracy to 5/5. - Committed the routing ground-truth rule and labelled evaluation set to version control before writing the router, so reported accuracy could not be fitted to observed behaviour; reported results as a range across five runs rather than a single best score.	
Source-Governed RAG <i>Python, FastAPI, ChromaDB, MiniLM/ONNX, PostgreSQL, Next.js · Render / Vercel / Neon</i> - Built an end-to-end RAG system over regulatory documents where a governance layer vetoes untrustworthy sources, with approval status and review dates overriding similarity ranking. - Improved retrieval accuracy from 78% to 95% on a labelled evaluation set by replacing fixed-size character chunking with document-structure chunking.	Live demo